

KeralaGO — Production Mobile Frontend Routing Specification

Complete Screen-by-Screen Navigation Inventory, RBAC Controls & Expo Router Architecture

Platform: React Native / Expo Router v4	App Root: app/src/app	Auth: JWT Bearer (SecureStore)	Roles: USER (Customer) DRIVER (Partner)
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1. Two-Tier Route Guards & 12-Step Ride Lifecycle

- **Authentication Guard:** Intercepts all requests at root _layout.tsx. If valid JWT in expo-secure-store is missing, forces redirect to (auth)/*.
- **Role-Based Access Control (RBAC):** USER is restricted to (customer)/*. DRIVER must pass KYC review (PENDING directs to (driver)/onboarding/*, APPROVED unlocks the duty console).
- **12-Step Core Lifecycle:** 01 Login → 02 Pickup & Drop → 03 Fare Quote → 04 Request → 05 Match → 06 Accept → 07 Driver Arrive → 08 Trip PIN + Start → 09 Live Tracking → 10 Complete → 11 Payment → 12 Bilateral Rating.

2. Master Screen Navigation & Authorization Inventory

Expo Route Path	Screen Name	Auth	Role Guard	Purpose, UI Components & Interactions
1. Authentication & Onboarding Flow ((auth))				
/index	Gatekeeper	Public	None	Initial splash; checks stored token and auto-redirects to customer or driver home
/(auth)/welcome	Welcome Carousel	Public	None	Onboarding walkthrough illustrating safety features, English/Malayalam language selector
/(auth)/login	Phone Login	Public	None	+91 country picker, 10-digit mobile number input, 'Send OTP' action button
/(auth)/otp-verify	OTP Verification	Public	None	4-digit OTP PIN auto-fill input boxes, 60s countdown timer, 'Resend Code' trigger
/(auth)/register-profile	Profile Setup	Post-OTP	None	First-time user onboarding: legal full name, optional email, and role selector (USER / DRIVER)
2. Customer / Passenger Application ((customer)) — Protected (USER)				
/(customer)/(tabs)/home	Map & Hail Home	Protected	USER	Interactive map with live moving cabs, 'Where do you want to go?' search, Suggestions: Ride, Bike, Auto, Pink Ride, Package, Share Taxi, Rentals, Schedule; Recent places
/(customer)/search-location	Search Location	Protected	USER	Dual inputs for Pickup and Destination, GPS 'Current Location' button, Google Places autocomplete
/(customer)/route-preview	Route Preview	Protected	USER	Route polyline display on map, pickup/drop pin fine-tuning, estimated distance & ETA
/(customer)/choose-ride	Choose a Ride	Protected	USER	Category selection: Car (■120-150, 4m), Bike (■40-60), Auto (■60-90), Share Taxi, Pink Ride (women-only); Payment method selector (Cash/UPI), Promo Code trigger, 'Confirm Ride' CTA
/(customer)/confirm-pickup	Confirm Pickup	Protected	USER	Fine-tune pickup marker on map, landmark instructions for driver
/(customer)/finding-driver	Driver Radar	Protected	USER	Pulsing radar animation, 3-step live status tracker (Searching -> Checking -> Confirming), cancel ride CTA
/(customer)/active-ride	Live Ride Tracking	Protected	USER	Live driver tracking, Driver Card (Photo, Rijin S, 4.8★, Maruti Swift - White, KL 07 AB 1234), Call/Chat buttons, 4-Digit Trip PIN display , ETA progress (8 min / 2.3 km), Share Trip, Safety SOS
/(customer)/trip-completed	Receipt & Rating	Protected	USER	Celebration badge, 'Trip Completed! ■132 Paid via UPI', View breakdown link, driver card, interactive 5★ rating widget, comment box, Done CTA
/(customer)/(tabs)/bookings	Bookings History	Protected	USER	Tabs: [Recent / Completed] vs [Scheduled]; Date-grouped trip cards with pickup, drop, fare, status badges
/(customer)/booking-details/[id]	Trip Receipt	Protected	USER	Itemized bill breakdown: base fare, distance, time, taxes, payment method, driver card, invoice download
/(customer)/schedule-ride	Schedule Ride	Protected	USER	Pickup and drop pickers, calendar date selector, time picker, repeat options (None/Daily/Weekly), Schedule CTA
/(customer)/(tabs)/profile	Customer Profile	Protected	USER	User header (Avatar, Devarth, Phone), Menu: Payment Methods, Addresses, Preferences, Safety & Support, Invite, Help, Logout
/(customer)/payment-methods	Payment Methods	Protected	USER	Saved options: UPI (GPay/PhonePe), Credit/Debit Cards, Cash, KeralaGo Wallet balance, 'Add New UPI ID' CTA
/(customer)/saved-addresses	Saved Addresses	Protected	USER	Manage Home, Work, and favorite custom destination locations
/(customer)/ride-preferences	Ride Preferences	Protected	USER	AC/Non-AC preference, quiet ride mode toggle, luggage handling assistance
/(customer)/safety-support	Safety & Support	Protected	USER	Safety hub: Share Live Location, Emergency SOS, Safety Toolkit guidelines, Report an Issue, Help & FAQs
/(customer)/emergency-contacts	Safety Contacts	Protected	USER	Manage verified contacts for emergency distress alerts with primary contact toggle
/(customer)/support-tickets	Support Tickets	Protected	USER	Dispute ticket creation form (category, ride picker, description) and open ticket history
3. Driver Partner Application ((driver)) — Protected (DRIVER)				
/(driver)/onboarding/license	License Upload	Driver KYC	DRIVER	Driving license number input, validity expiry date, front and back document photo upload
/(driver)/onboarding/vehicle	Vehicle Details	Driver KYC	DRIVER	Vehicle registration plate, make, model, year, category selection, seating capacity
/(driver)/onboarding/rc-insurance	RC & Insurance	Driver KYC	DRIVER	Registration Certificate (RC Book) and active Insurance policy document upload
/(driver)/onboarding/bank-details	Bank & Payouts	Driver KYC	DRIVER	Bank account number, IFSC code, account holder name, and UPI ID for earnings disbursement
/(driver)/onboarding/status	KYC Review Status	Driver KYC	DRIVER	Under review status screen or rejection notice with administrator feedback

Expo Route Path	Screen Name	Auth	Role Guard	Purpose, UI Components & Interactions
<code>/(driver)/(tabs)/home</code>	Driver Duty Console	Protected	Approved	Online/Offline toggle switch , Today's stats: 12 Trips, █1,240 Earnings, 4.8 Rating; Peak hours alert banner; Upcoming opportunities with navigate button
<code>/(driver)/incoming-request</code>	Ride Offer Overlay	Protected	Approved	Full-screen 20-second circular countdown, '5 min away', Pickup (M B Hostel), Drop (MACE), 2.8 km, █120 fare, big green 'Accept' and dark 'Decline' buttons
<code>/(driver)/arriving-pickup</code>	Navigation to Pickup	Protected	Approved	Turn-by-turn banner '200m Turn left onto MACE Hostels Rd', map route to pickup, 'Arriving at pickup 2 min • 800m', Customer card (Devarth, 4.8★), Call/Chat, 'Arrived at Pickup' CTA
<code>/(driver)/pickup-passenger</code>	Pickup Passenger	Protected	Approved	'Pickup Passenger - Rider is waiting', customer card, 'Start Trip' slider prompting 4-digit customer PIN verification modal
<code>/(driver)/in-trip</code>	Trip in Progress	Protected	Approved	Turn-by-turn navigation header '1.2 km Continue straight', destination route, live speed, Trip stats: 12 min left, 5.6 km, █120 fare est., big red 'End Trip' button
<code>/(driver)/trip-summary</code>	Trip Completed	Protected	Approved	'Trip Completed █120 Your Earnings', breakdown: Base Fare █100, Distance (5.6 km) █20; Rate Rider card (5 interactive stars, comment), Done button
<code>/(driver)/(tabs)/earnings</code>	Earnings Dashboard	Protected	Approved	Filters: [Daily, Weekly, Monthly]; Total: █1,240, 12 Trips, 10h 20m online, █103/trip; Hourly bar chart (6, 9, 12, 15, 18, 21h); Breakdown: Trip █1,180, Tips █60; 'Request Payout' button
<code>/(driver)/payout-request</code>	Payout Withdrawal	Protected	Approved	Available balance card, withdrawal amount input, bank account selector, 'Withdraw to Bank' action CTA
<code>/(driver)/(tabs)/trips</code>	Driver Trips	Protected	Approved	Filter tabs: [Completed / Cancelled]; Daily grouped trip cards with time, route, and earnings (█120, █85, █165, █110)
<code>/(driver)/trip-details/[id]</code>	Driver Trip Audit	Protected	Approved	Itemized statement: Gross fare, platform commission deducted, net credited to ledger, rider details
<code>/(driver)/(tabs)/profile</code>	Driver Profile	Protected	Approved	Profile card: Photo, Rijin S, Verified badge, 4.8★, 320 trips, 6 months; Menu: Personal Info, Vehicle Details, Documents [Verified], Bank Account, Notifications, Language, Help, Logout
<code>/(driver)/vehicle-details</code>	Vehicle Details	Protected	Approved	Car photo (Maruti Swift - White, KL 07 AB 1234), Model, Color, Year 2022, Verification badges: RC Verified, Insurance Verified, PUC Verified
<code>/(driver)/documents-management</code>	Documents Hub	Protected	Approved	List of compliance documents with expiry dates, verification badges, and re-upload actions
<code>/(driver)/safety-support</code>	Driver Safety	Protected	Approved	Emergency SOS, Share Live Location, Safety Guidelines, Report an Issue, Contact Support
<code>/(driver)/settings</code>	Driver Settings	Protected	Approved	Toggles: Go Online Automatically, Request Sound; Navigation App picker (Google Maps), Distance Unit (km), Appearance (Light/Dark), Privacy, Terms, v1.0.0
4. Global Modal Overlays ((modals)) — Cross-Role				
<code>/(modals)/sos</code>	Emergency SOS	Protected	Any Role	One-tap emergency broadcast to backend, live location sent to emergency contacts, emergency dialer trigger (112)
<code>/(modals)/chat/[rideId]</code>	In-Ride Chat	Protected	Participants	Real-time in-app WebSocket messaging thread between rider and assigned driver during an active trip
<code>/(modals)/cancel-ride/[id]</code>	Cancel Dialog	Protected	Participants	Standardized reason list ('Driver took too long', 'Changed plans', 'Driver asked cash') with cancellation fee notice
<code>/(modals)/verify-pin</code>	Verify PIN Modal	Protected	Driver	4-digit input keypad for driver to validate passenger's start PIN before beginning trip
<code>/(modals)/promo-code</code>	Promo Code	Protected	USER	Coupon code input sheet with discount application preview

3. Production Frontend Engineering Checklist

- **Secure Token Storage:** Store JWT Access and Refresh tokens in expo-secure-store with automatic refresh interceptors.
- **Real-Time WebSocket Stream:** Single persistent socket connection handling driver GPS telemetry and instant ride offers.
- **Background Geolocation:** Continuous driver GPS stream via expo-location + expo-task-manager while screen locked.
- **Push Notifications:** Firebase Cloud Messaging (FCM) integration with deep link route forwarding for ride acceptance/completion.
- **Offline Resilience:** Persistent React Query cache for trip receipts, driver earnings history, and emergency contacts.